

QUEBEC CITY SAINTE-FOY, QC, Canada

WMO: 713920

Lat: 46.7803N

Lon: 71.2875W

Elev: 91

StdP: 100.23

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -24.0	(c) -21.5	(d) -29.4	(e) 0.3	(f) -23.2	(g) -26.7	(h) 0.3	(i) -20.9	(j) 10.4	(k) -7.9	(l) 9.4	(m) -6.8	(n) 2.8	(o) 250	(p) 0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	29.1	20.6	27.5	19.7	26.0	18.9	22.4	26.7	21.4	25.2	20.5	24.1	3.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 21.1	(b) 15.9	(c) 24.5	(d) 20.1	(e) 15.0	(f) 23.7	(g) 19.1	(h) 14.0	(i) 22.6	(j) 66.4	(k) 26.8	(l) 62.6	(m) 25.2	(n) 59.4	(o) 24.2	(p) 27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
9.1	7.8	6.5	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-27.2	31.9	3.0	1.6	-29.3	33.0	-31.1	33.9	-32.7	34.8	-34.9	35.9
			-27.9	24.2	3.0	1.2	-30.0	25.1	-31.8	25.8	-33.4	26.5	-35.6	27.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.6	-10.8	-8.7	-3.5	4.2	12.2	16.9	20.1	19.2	14.8	7.5	0.9	-6.5		
	DBStd	11.67	6.60	5.79	5.60	4.20	4.15	3.59	2.83	2.87	3.81	3.95	4.50	5.80		
	HDD10.0	2675	644	524	419	179	23	1	0	0	4	97	274	510		
	HDD18.3	4810	902	758	677	424	196	68	14	24	119	335	523	768		
	CDD10.0	1069	0	0	0	4	90	209	312	284	148	21	1	0		
	CDD18.3	162	0	0	0	0	5	26	68	51	12	0	0	0		
	CDH23.3	1119	0	0	0	3	71	203	454	305	83	1	0	0		
	CDH26.7	229	0	0	0	0	14	48	98	54	14	0	0	0		
(14) Wind	WSAvg	2.8	3.0	3.1	3.1	3.3	3.2	2.6	2.3	2.2	2.2	2.7	2.8	2.9		
(15) Precipitation	PrecAvg	1266	96	85	91	82	102	114	123	121	127	99	112	117		
	PrecMax	1648	159	192	150	154	196	182	200	224	218	171	184	221		
	PrecMin	988	43	6	21	22	32	40	57	33	36	37	50	47		
	PrecStd	148	35	46	29	29	44	35	38	53	41	34	34	36		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.4	5.1	11.5	20.4	28.6	30.4	31.4	30.2	28.5	21.6	14.7	6.6		
		MCWB	4.6	3.5	6.4	12.3	17.5	20.9	22.7	21.9	21.2	16.3	11.9	5.8		
	2%	DB	2.8	2.6	7.5	16.4	25.2	28.0	29.2	28.2	25.7	18.2	10.7	3.9		
		MCWB	2.0	1.0	3.7	8.8	15.7	19.3	21.2	20.4	19.6	14.0	8.5	2.9		
	5%	DB	0.8	1.4	5.2	13.1	22.5	25.6	27.5	26.5	23.4	15.4	8.6	2.4		
		MCWB	0.2	0.3	2.5	7.2	14.8	18.1	20.0	19.6	18.3	12.6	6.7	1.4		
	10%	DB	-1.4	0.0	3.4	10.6	19.7	23.5	25.9	24.9	21.2	13.3	6.7	1.0		
		MCWB	-2.2	-1.0	1.3	5.8	13.4	17.1	19.1	18.9	17.1	10.8	5.0	0.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	3.7	8.1	14.3	20.1	22.6	24.1	23.2	22.5	17.4	12.3	5.9		
		MCDB	4.8	4.6	10.2	19.4	25.4	28.0	29.2	28.2	26.6	20.0	13.9	6.4		
	2%	WB	2.0	1.4	4.5	9.9	18.1	21.0	22.7	22.0	20.8	14.9	9.2	3.2		
		MCDB	2.6	2.3	6.6	13.6	22.8	25.6	26.9	25.8	24.3	17.3	10.4	3.7		
	5%	WB	0.1	0.4	2.9	8.1	15.9	19.8	21.5	21.1	19.2	13.1	7.0	1.6		
		MCDB	0.8	1.2	4.8	12.4	20.0	23.8	25.3	24.6	22.3	15.3	8.3	2.2		
	10%	WB	-2.1	-1.1	1.5	6.5	14.0	18.6	20.5	20.1	17.6	11.2	5.4	0.4		
		MCDB	-1.5	-0.2	3.1	10.0	18.9	22.4	24.0	23.6	20.7	13.0	6.7	1.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	8.5	8.2	8.2	9.9	9.7	9.4	9.4	9.4	7.1	5.9	6.3		
		MCDBR	8.2	8.3	9.7	13.5	14.7	12.8	11.6	11.6	11.9	10.6	8.8	6.4		
		MCWBR	8.1	7.2	7.0	8.2	7.5	6.4	5.5	5.5	6.5	7.3	7.2	6.3		
		MCWBR	8.2	7.5	8.9	11.5	12.3	10.8	9.8	9.4	10.4	9.6	8.2	6.2		
(40) Clear-Sky Solar Irradiance	taub	taub	0.264	0.283	0.325	0.350	0.365	0.402	0.412	0.394	0.368	0.337	0.307	0.292		
		taud	2.316	2.249	2.221	2.348	2.398	2.291	2.283	2.336	2.401	2.481	2.471	2.278		
		Ebn at Noon	834	887	895	905	900	864	849	851	843	820	781	738		
		Edh at Noon	74	98	118	116	114	128	127	116	99	79	65	70		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.41	2.31	3.52	4.44	5.06	5.52	5.41	4.70	3.69	2.15	1.25	0.95		
		RadStd	0.20	0.28	0.29	0.38	0.53	0.35	0.53	0.29	0.31	0.18	0.15	0.09		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page